

Motivation

HOA microphones are validated almost exclusively with **anechoic test signals**. Do systematic differences survive in a **real concert hall** with **live piano**?

A **Zylia ZM-1** and a **Harpex Spcmic** were mounted side-by-side (17 cm apart) in a 370-seat concert hall. Four metrics were extracted from B-format files.

Microphone Arrays — August 2024



Harpex Spcmic (centre), ZM-1 under test (left). The right ZM-1 was part of a separate concurrent subtest, not included in this comparison.

Concert Hall Stage — August 2024



Aula overview. RT60 = 1.97 s, capacity 370 seats.

Recording Setup — April 2024

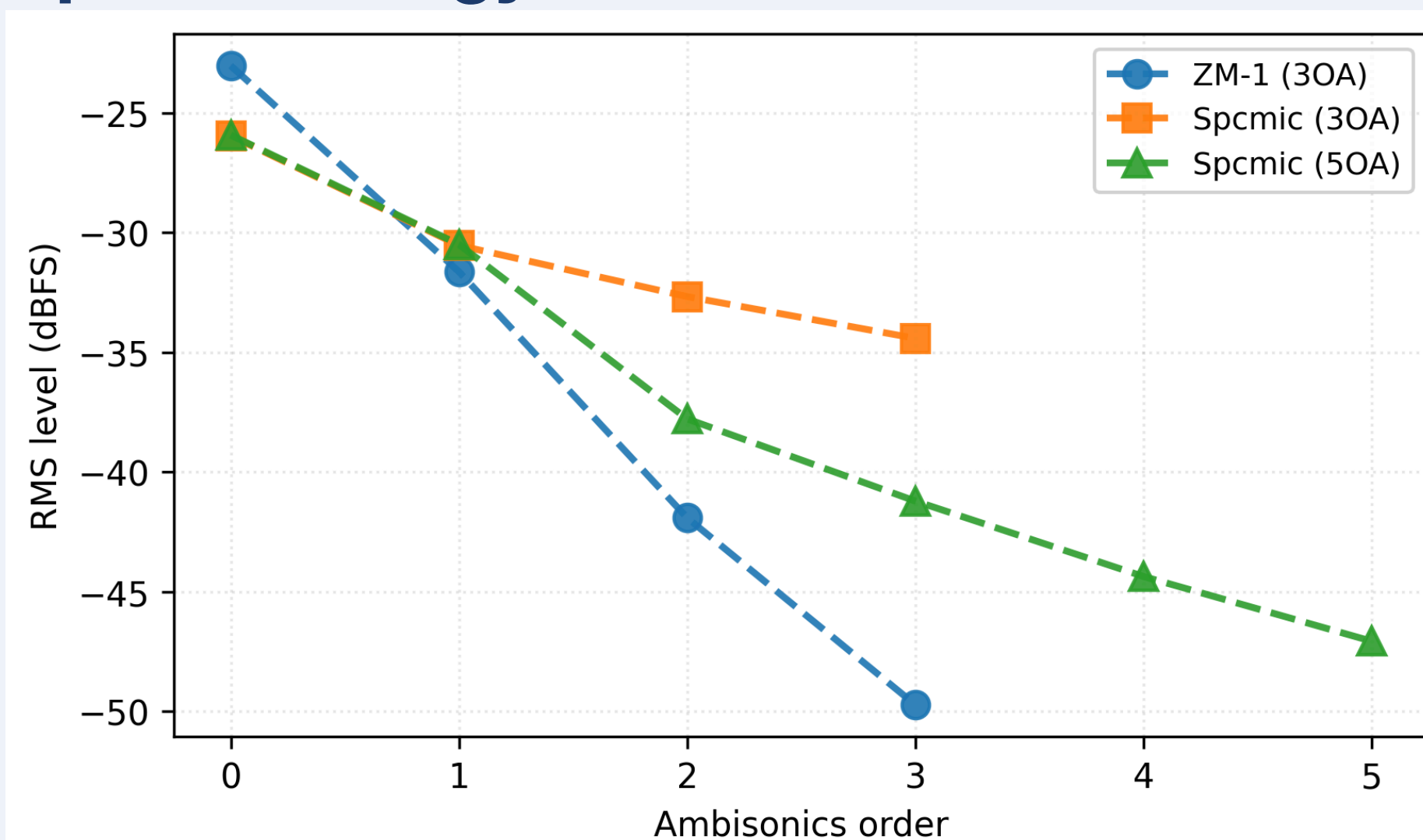


Gdańsk Univ. of Technology Main Aula. Arrays 1.5 m from Steinway D.

19 dB spatial-energy rolloff gap

At **nominally identical 3rd-order**, the ZM-1 attenuates **27.4 dB** from order 0→3; Spcmic 3OA attenuates only **8.4 dB** — consistent across all repertoire and **below the spatial-aliasing limit**.

Spatial Energy Distribution

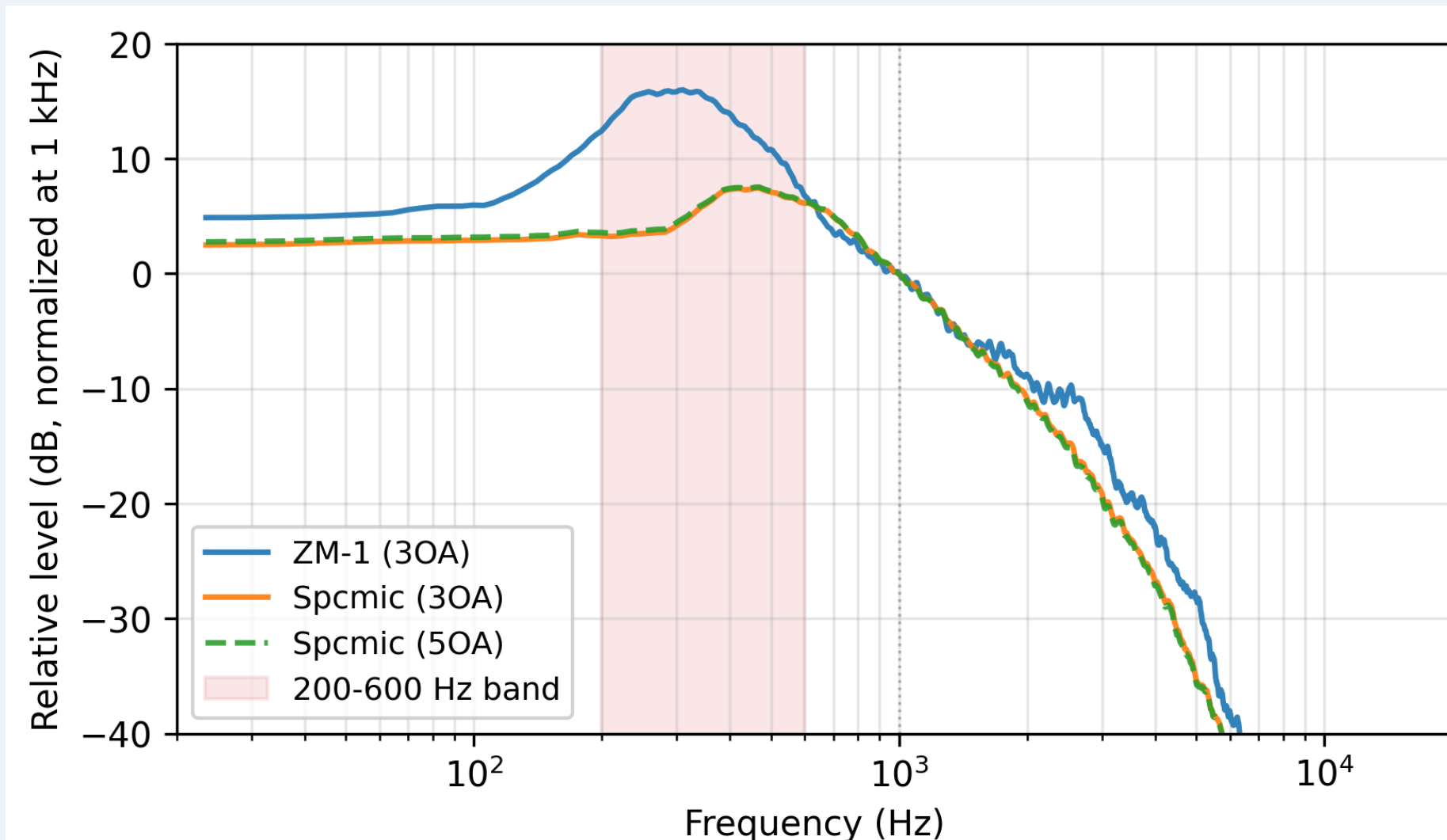


Per-order dBFS (RMS-mean per channel). August session. Both labelled “3OA” — 19 dB rolloff gap between orders 0 and 3.

Arrays Under Test

	ZM-1	Spcmic
Geometry	spherical	planar
Diameter	103 mm	170 mm
Capsules	19 MEMS	84 MEMS
Amb. order	3rd	3rd / 5th
Encoding	Zylia Conv.	Spcmic app / Spacemic VST

W-Channel Spectral Response



ZM-1 shows 5.2–11 dB elevation at 200–600 Hz (mean 9.1 dB) — possibly related to near-field diffraction or capsule frequency response of the compact spherical enclosure.

Recording Sessions

Primary comparison (Aug 2024)

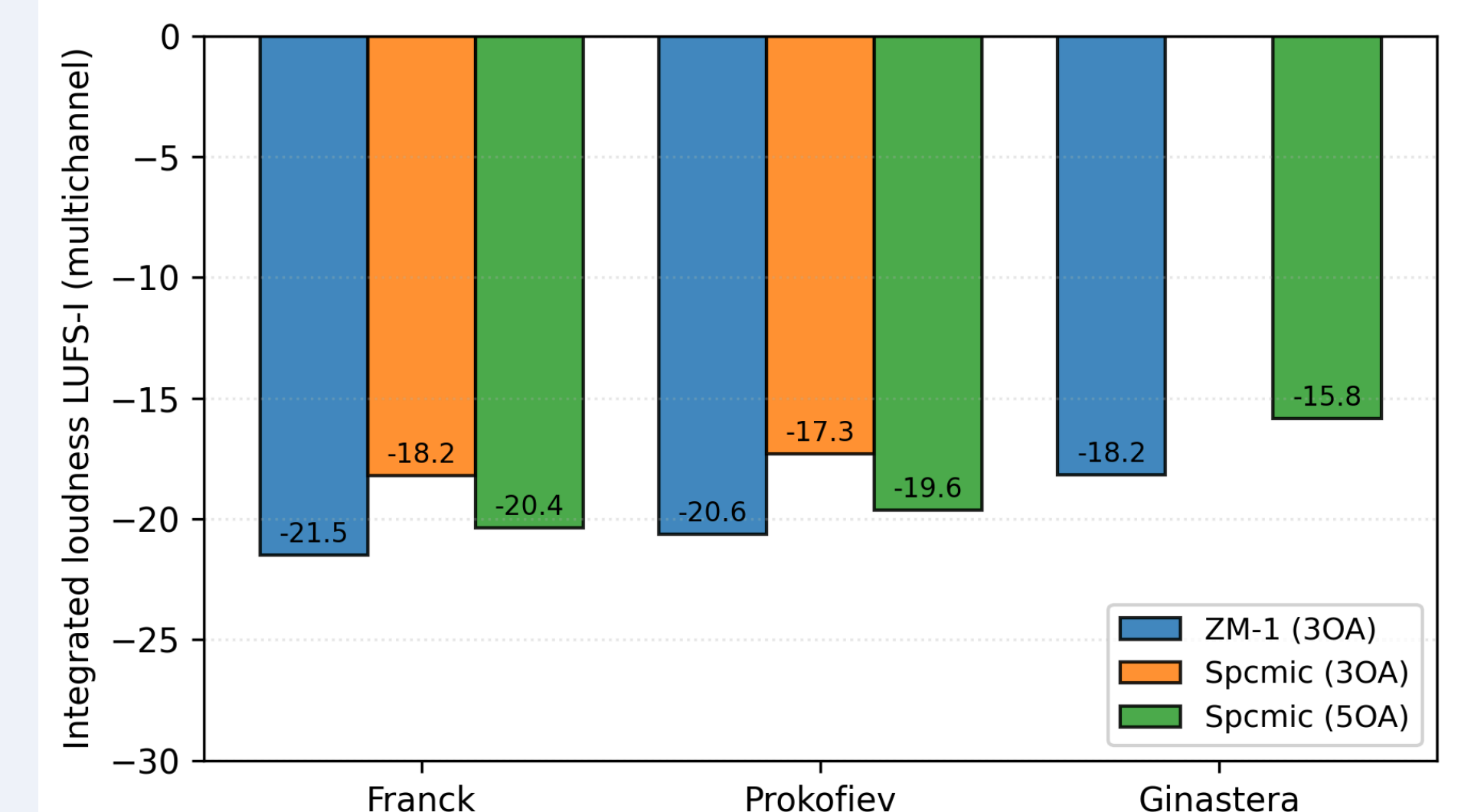
→ Franck + Prokofiev; 35.5 min
→ ZM-1 & Spcmic 3OA + 5OA

Consistency check (Apr 2024)

→ Ginastera; 15.4 min
→ ZM-1 & Spcmic 5OA only

Both normalised to −0.5 dBTP; ambiX/SN3D.

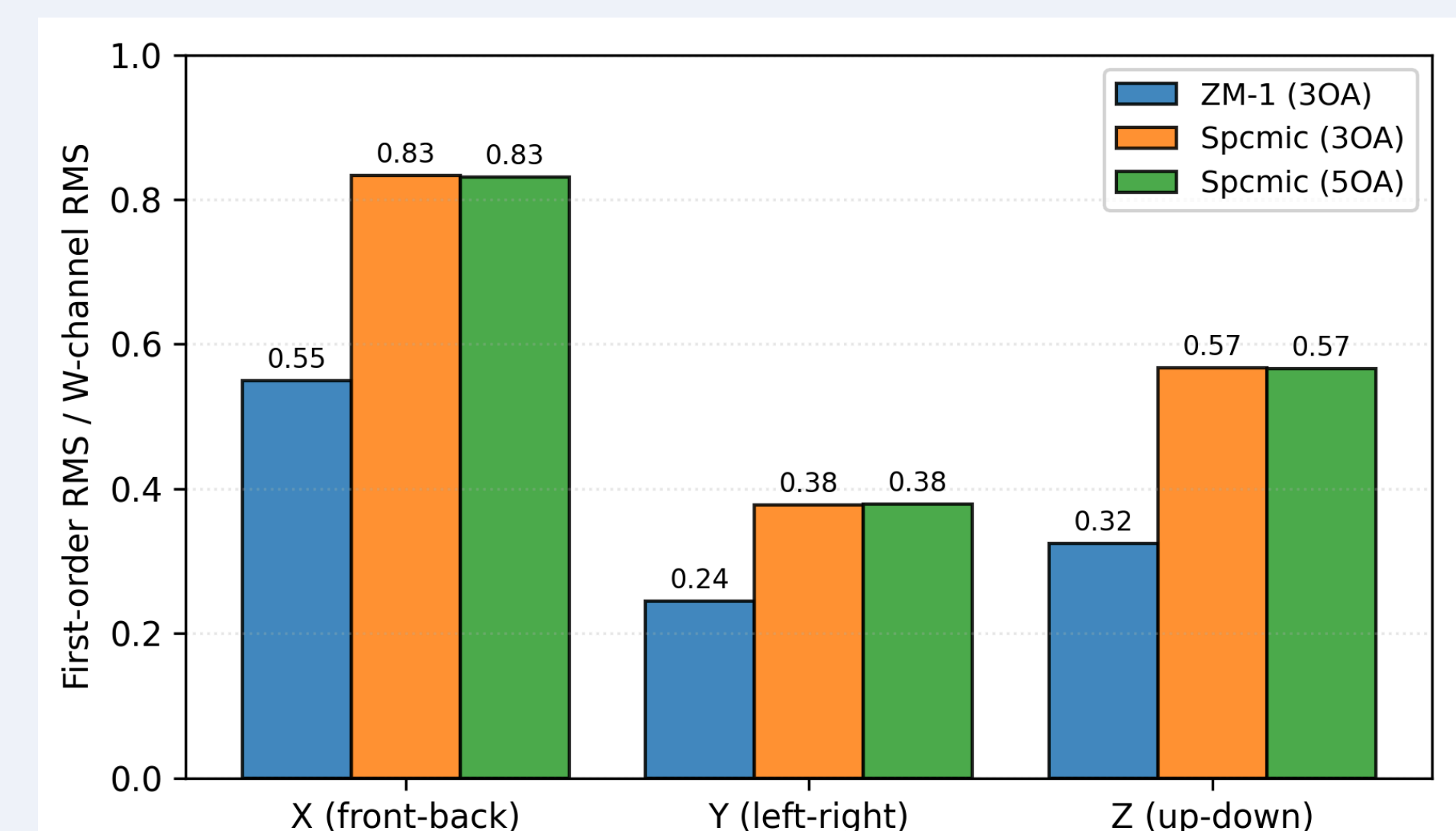
LUFS-I (Integrated Loudness)



Spcmic 2.3–3.3 dB louder (multichannel). **W-channel alone inverts**: ZM-1 is 0.8–2.6 dB louder — higher-order components carry the difference.

1st-Order Directional Ratios

X/W: ratio of 1st-order (X) to omnidirectional (W) RMS energy; 1.0 = ideal dipole.



Spcmic X/W ≈ 0.83 vs. ZM-1 ≈ 0.55 — consistent with gentler energy rolloff of planar geometry.

Sub-Aliasing Confirmation

Restricting to **f < 3 kHz** (SPARTA array2sh ZM-1 aliasing limit) still yields a **19.3 dB** rolloff gap — the result is **not an aliasing artefact**.

Summary

- **Nominal order ≠ per-order energy**. Two “3OA” arrays differ by 19 dB.
- **Geometry drives rolloff**. Planar → gentle; compact spherical → steep.
- **Loudness inverts** on W-channel vs. multi-channel.
- Next step: MUSHRA perceptual evaluation.

Data & Reproducibility

Recordings DOI

(CC BY-NC-SA 4.0):
10.34808/5xe2-ah94

Analysis pipeline & full paper

(CC BY 4.0):
git.pg.edu.pl/p829296/
hoa-mic-comparison-aes2026/

